DATA SCIENCE PROJECT

House Price Prediction using Regularized Regression

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A Comparative Analysis of Ridge and Lasso Regression on the Boston Housing Dataset

506 Housing Records

13 Predictive Features

2 Regularization Models

PRESENTATION OVERVIEW

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Project Overview & Goals

Primary Objectives

- **Understand Feature-Price Relationships:** Analyze how socio-economic, environmental, and structural attributes influence house prices
- **Handle Multicollinearity:** Apply regularization techniques to address correlated features without manual removal
- **Compare Regularization Models:** Build and evaluate Ridge (L2) and Lasso (L1) regression approaches
- **Optimize Model Selection:** Use validation-based approach to select optimal regularization strength (λ)
- **Evaluate Performance:** Assess models using MAE, RMSE, and MAPE metrics

Tools & Libraries

Python | Pandas | NumPy | Matplotlib | Seaborn | Scikit-learn | Statsmodels

Project Workflow

1. **Data Preparation**
   Load dataset, check missing values, split into train/validation/test sets
2. **Exploratory Analysis**
   Visualize distributions, identify skewed variables, analyze feature relationships
3. **Correlation Analysis**
   Create correlation heatmap, detect multicollinearity using VIF scores
4. **Model Training**
   Train Ridge and Lasso models, tune λ parameter using validation set
5. **Evaluation**
   Compare performance using MAE, RMSE, MAPE metrics

DATA FOUNDATION

Dataset Overview: Boston Housing Data

Feature Descriptions

crim:	Crime rate per town	**zn:**	Proportion of large residential land
indus:	Proportion of industrial area	**chas:**	Proximity to Charles River (0/1)
nox:	Nitric oxide concentration	**rm:**	Average rooms per dwelling
age:	Proportion of older buildings	**dis:**	Distance to employment centers
rad:	Accessibility to major highways	**tax:**	Property tax rate
ptratio:	Student-teacher ratio	**black:**	Demographic indicator
lstat:	Percentage of lower-status population		
medv:	**Median house price (TARGET)**		

Total Records
506

Features
13

Target Range
$5K-$50K

Data Split Strategy

Training Set — **60%**

304 samples for model training

Validation Set — **20%**

101 samples for hyperparameter tuning

Test Set — **20%**

101 samples for final evaluation

Feature Categories

- **Socio-economic:** crim, lstat, black
- **Environmental:** nox, indus
- **Structural:** rm, age, zn
- **Accessibility:** dis, rad, chas
- **Fiscal:** tax, ptratio

DATA QUALITY ASSESSMENT

Exploratory Data Analysis: Feature Distributions

Distribution Patterns

Right-Skewed Variables

CRIM, ZN, LSTAT — Majority of areas show low values with few extreme outliers. CRIM indicates most neighborhoods have low crime rates; LSTAT reflects socioeconomic stratification.

Approximately Normal

RM, MEDV — Bell-shaped distributions indicate consistent room counts and price ranges across dwellings, suitable for linear modeling assumptions.

Discrete Variables

CHAS, RAD — Binary and categorical distributions representing proximity to Charles River and highway accessibility indices.

Left-Skewed

AGE — High concentration of older buildings, indicating mature neighborhoods with established housing stock.

Data Quality Summary

- ✓ **No Missing Values**
  All 506 records complete
- ✓ **No Duplicates**
  Unique housing records
- ✓ **Outliers Retained**
  Represent real conditions
- ✓ **Consistent Distributions**
  Train/test sets aligned

Outlier Handling Rationale

Limited Dataset Size

With only 506 total records (404 train, 102 test), removing outliers would significantly reduce sample size and potentially introduce bias.

Real-World Representation

Extreme values represent genuine housing market conditions: high-crime areas, luxury properties, industrial zones, and premium neighborhoods.

Regularization Robustness

Ridge and Lasso regression are inherently robust to outliers through their penalty mechanisms, making outlier removal unnecessary.

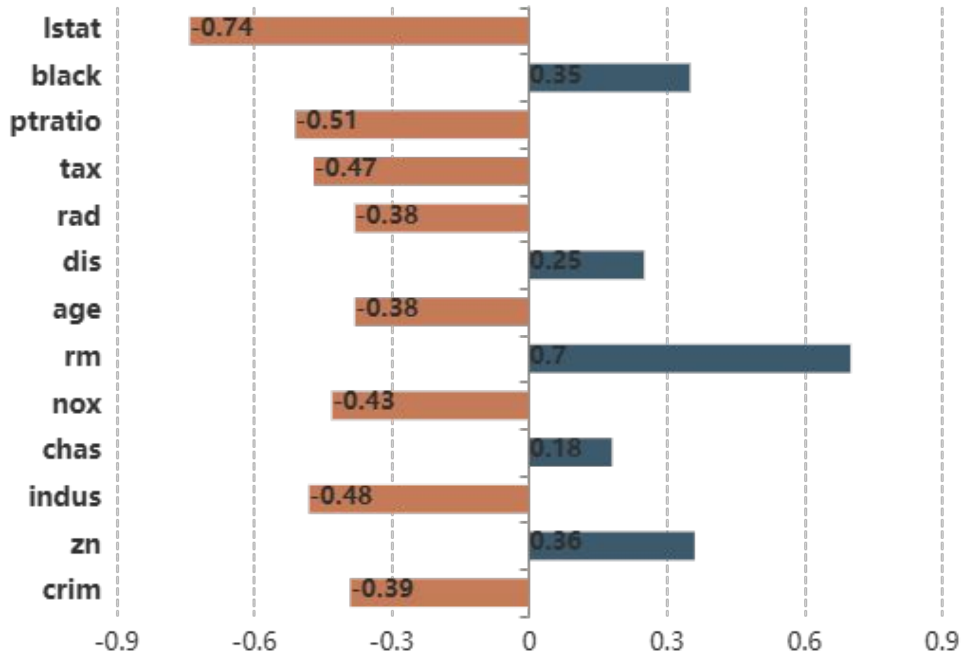
Consistent Patterns

Outliers appear consistently in both training and testing sets, confirming they are not data entry errors but genuine market variations.

FEATURE RELATIONSHIPS

Correlation Analysis & Multicollinearity Detection

Feature Correlations with House Price (medv)



↑ Strongest Positive

rm (rooms)	+0.70
black	+0.35
chas	+0.18

↓ Strongest Negative

lstat	-0.74
ptratio	-0.51
indus	-0.48

Multicollinearity Detection (VIF Analysis)

tax Property tax rate	9.01 Highest VIF
rad Highway accessibility	7.45 Correlated with tax
nox Nitric oxide conc.	6.29 Environmental factor
indus Industrial proportion	5.89 Land use factor
age Building age	4.87 Structural factor

VIF Threshold: Values < 10 indicate acceptable multicollinearity levels. Regularization will handle remaining correlations without manual feature removal.

REGULARIZATION TECHNIQUES

Ridge Regression: L2 Regularization

Mathematical Formulation

$$J(\beta) = MSE + \lambda \sum \beta^2$$

$J(\beta)$: Cost function to be minimized

MSE: Mean Squared Error between predicted and actual values

λ (lambda): Regularization parameter controlling penalty strength

$\sum \beta^2$: L2 penalty: sum of squared coefficient magnitudes

How Ridge Works

1. Adds penalty proportional to the **square of coefficient magnitudes**
2. Shrinks all coefficients **smoothly toward zero** but never exactly to zero
3. Distributes effects among **correlated features** rather than selecting one
4. Larger λ = more shrinkage = simpler model = potentially higher bias

Key Strengths

Multicollinearity Handling

Effectively manages highly correlated features by distributing coefficient values among them, preventing any single feature from dominating the model.

Model Stability

Produces more stable and reliable predictions compared to ordinary least squares when features are correlated, reducing variance in coefficient estimates.

Feature Retention

Retains all features in the model with reduced coefficients, preserving complete information and avoiding arbitrary feature elimination.

When to Use Ridge

- ✓ You believe **all features** have some meaningful relationship to the target
- ✓ Your dataset exhibits **multicollinearity** among predictors
- ✓ **Prediction accuracy** is the primary goal over interpretability
- ✓ You want a **stable, robust model** with lower variance

REGULARIZATION TECHNIQUES

Lasso Regression: L1 Regularization & Feature Selection

Mathematical Formulation

$$J(\beta) = MSE + \lambda \sum |\beta|$$

$J(\beta)$: Cost function to be minimized

MSE: Mean Squared Error between predicted and actual values

λ (lambda): Regularization parameter controlling penalty strength

$\sum|\beta|$: L1 penalty: sum of absolute coefficient values

How Lasso Works

1. Adds penalty proportional to the **absolute value of coefficients**
2. Can force some coefficients to be **exactly zero**, performing feature selection
3. Creates **sparse models** by keeping only the most important features
4. Larger λ = more zeros = simpler model = automatic feature selection

Key Strengths

Automatic Feature Selection

Inherently performs feature selection by eliminating irrelevant variables, creating interpretable models that identify only the most impactful predictors.

Model Simplicity

Produces sparse, parsimonious models that are easier to interpret and explain to stakeholders, especially valuable in high-dimensional datasets.

Dimensionality Reduction

Naturally reduces model complexity by zeroing out coefficients, effectively performing embedded dimensionality reduction without separate preprocessing.

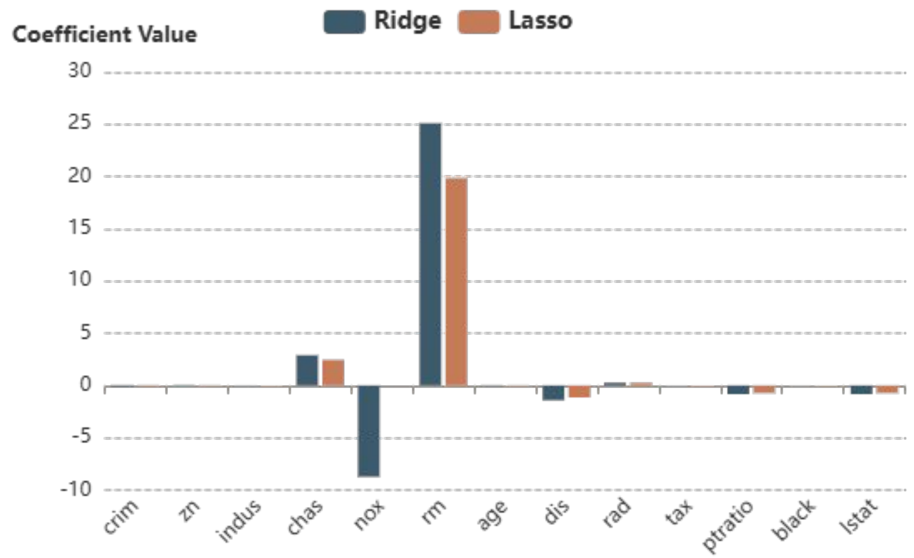
When to Use Lasso

- ✓ You suspect only a **subset of features** are truly important
- ✓ **Model interpretability** is a priority over raw accuracy
- ✓ You have a **high-dimensional dataset** with many features
- ✓ You need to **identify key drivers** for business insights

MODEL DEVELOPMENT

Model Training & Implementation

Coefficient Comparison: Ridge vs Lasso



Ridge Coefficients

All 13 features retained with reduced magnitudes

rm (rooms)	+25.10
lstat	-0.83
ptratio	-0.83

Lasso Coefficients

Feature selection: some coefficients = 0

rm (rooms)	+19.86
lstat	-0.73
nox	0.00

Training Methodology

1. Data Preprocessing

Load Boston dataset, verify data quality, split into 60/20/20 train/validation/test sets

2. Hyperparameter Tuning

Test multiple λ values (alpha), select optimal using validation RMSE

3. Model Training

Fit Ridge (λ=1) and Lasso (λ=0.01) using scikit-learn on training data

4. Coefficient Extraction

Extract intercept and feature coefficients for interpretation and comparison

🔑 Consistent Findings

Primary Positive & Negative Driver

rm (room count) shows strongest positive impact in both models, confirming larger homes command premium prices, while **lstat (lower-status %)** consistently shows strong negative impact, reflecting neighborhood socioeconomic effects

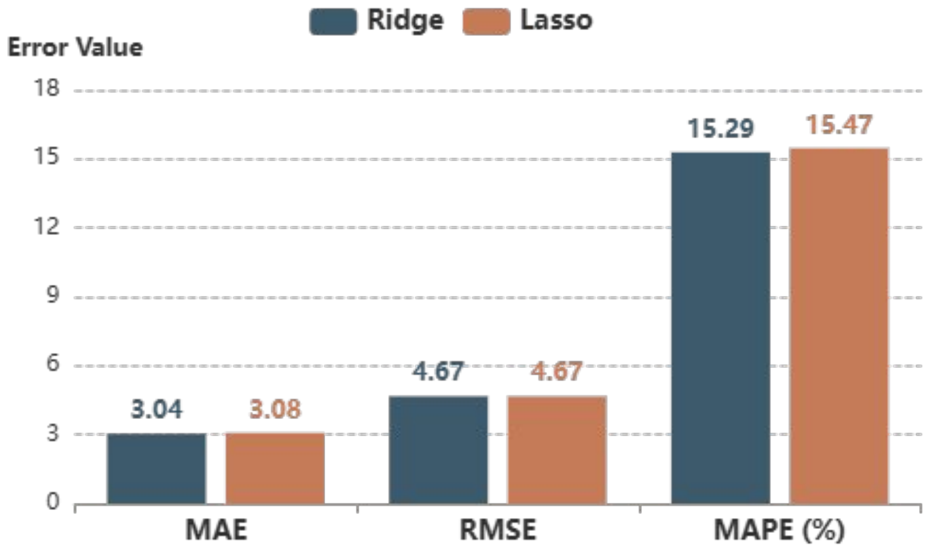
Model Agreement

Both models identify same key drivers, validating results align with housing economics theory

PERFORMANCE ASSESSMENT

Model Evaluation Results

Performance Metrics Comparison



Ridge MAE
3.04
Mean Absolute Error

Ridge RMSE
4.67
Root Mean Squared Error

Ridge MAPE
15.29%
Mean Absolute % Error

Detailed Results Analysis

Ridge Regression Performance

3.04	4.67	15.29%
MAE	RMSE	MAPE

Achieves slightly lower MAE and MAPE, indicating better overall prediction accuracy and relative error performance.

Lasso Regression Performance

3.08	4.67	15.47%
MAE	RMSE	MAPE

Nearly identical RMSE shows comparable robustness to outliers, with marginally higher error metrics.

Key Observations

- ✓ **Similar RMSE** values indicate both models handle outliers comparably
- ✓ **Small differences** suggest both models are well-calibrated
- ✓ **Ridge advantage** is marginal but consistent across metrics

ANALYTICAL INSIGHTS

Key Insights & Feature Interpretation

↑ Primary Price Drivers (Positive Impact)

rm (Average Rooms) +0.70

Strongest positive correlation. Each additional room significantly increases house value, reflecting the direct relationship between living space and price. Larger homes command premium prices in the Boston market.

black (Demographic) +0.35

Moderate positive correlation indicating socioeconomic factors. Higher values reflect better neighborhood conditions and community stability, contributing to property value appreciation.

chas (Charles River) +0.18

Binary variable showing proximity to Charles River adds modest premium. Waterfront access and scenic views contribute to desirability and higher valuations.

↓ Primary Price Drivers (Negative Impact)

lstat (Lower-Status %) -0.74

Strongest negative correlation. Higher percentage of lower-status population significantly depresses house prices, reflecting neighborhood socioeconomic challenges and perceived investment risk.

ptratio (Student-Teacher) -0.51

Strong negative impact. Higher student-teacher ratios indicate lower education quality, reducing neighborhood attractiveness for families and depressing property values.

indus (Industrial %) -0.48

Moderate negative correlation. Higher industrial presence reduces residential desirability due to noise, pollution, and traffic, negatively impacting housing demand and prices.

Validation with Housing Economics Theory

All findings align with established real estate principles: property size (rm), neighborhood quality (lstat, ptratio), environmental factors (indus, nox), and location amenities (chas) are well-documented price determinants. Model results demonstrate both statistical validity and domain coherence.

PROJECT SUMMARY

Conclusions & Recommendations

Model Selection Guidance

Choose Ridge Regression When:

- **Prediction accuracy** is the primary objective
- All features carry **meaningful information**
- You need **stable, robust predictions**
- Multicollinearity is present in the dataset

Choose Lasso Regression When:

- **Model interpretability** is critical
- You need **feature selection** to identify key drivers
- Working with **high-dimensional data**
- Creating **simpler, explainable models**

Key Takeaways

1. **Regularization Effectiveness**
   Both Ridge and Lasso successfully handle multicollinearity without manual feature removal, demonstrating the power of regularization techniques in real-world modeling.
2. **Consistent Feature Importance**
   Room count (rm) and lower-status percentage (lstat) emerge as primary price drivers in both models, validating results against housing economics theory.
3. **Domain Knowledge Integration**
   Combining statistical insights with real estate domain understanding enhances model trustworthiness and practical applicability.
4. **Model Performance**
   Ridge achieves marginally better accuracy (MAE: 3.04 vs 3.08), while Lasso provides simpler, more interpretable models through feature selection.

Future Directions

Consider exploring **Elastic Net regression** (combining Ridge and Lasso penalties) to leverage both stability and feature selection. Additionally, non-linear models (Random Forest, XGBoost) could capture complex interactions between features.